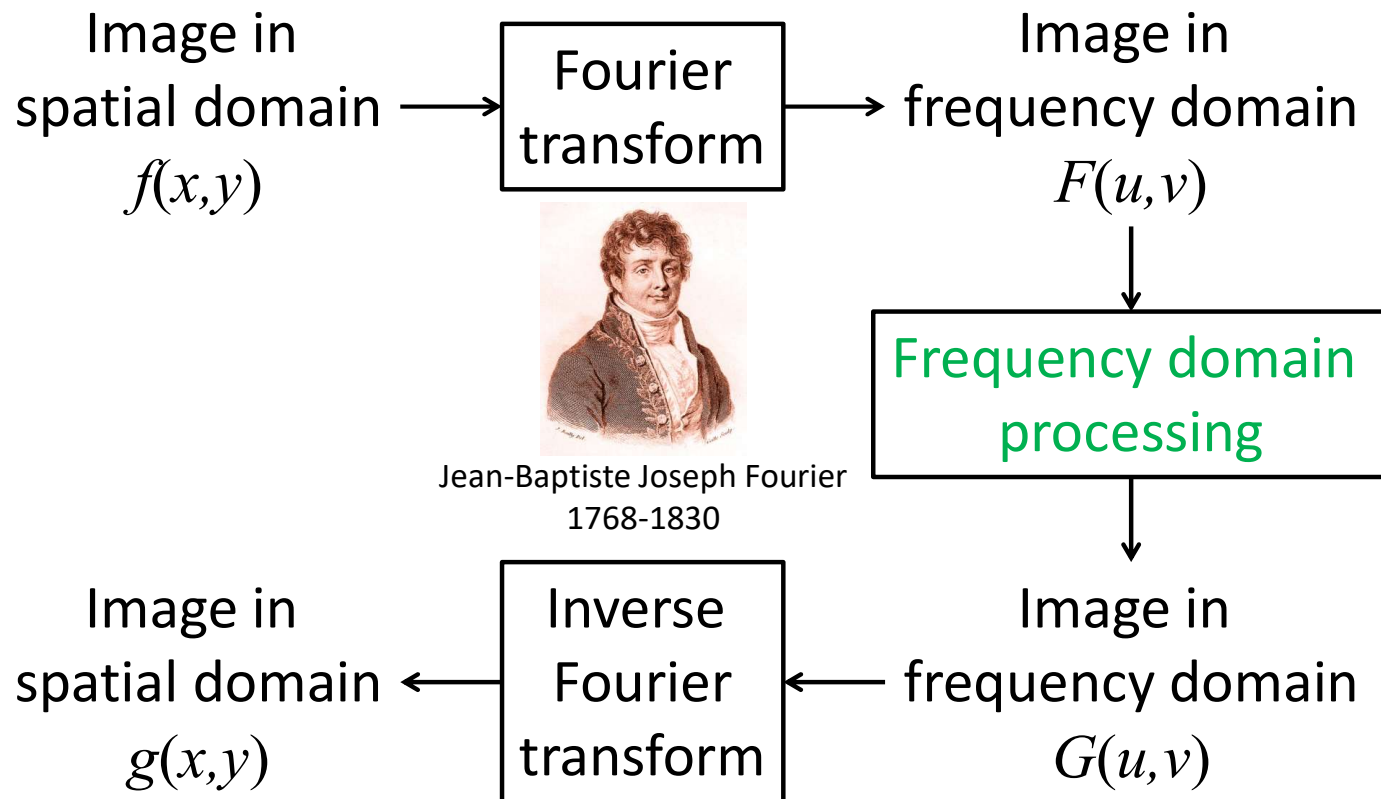


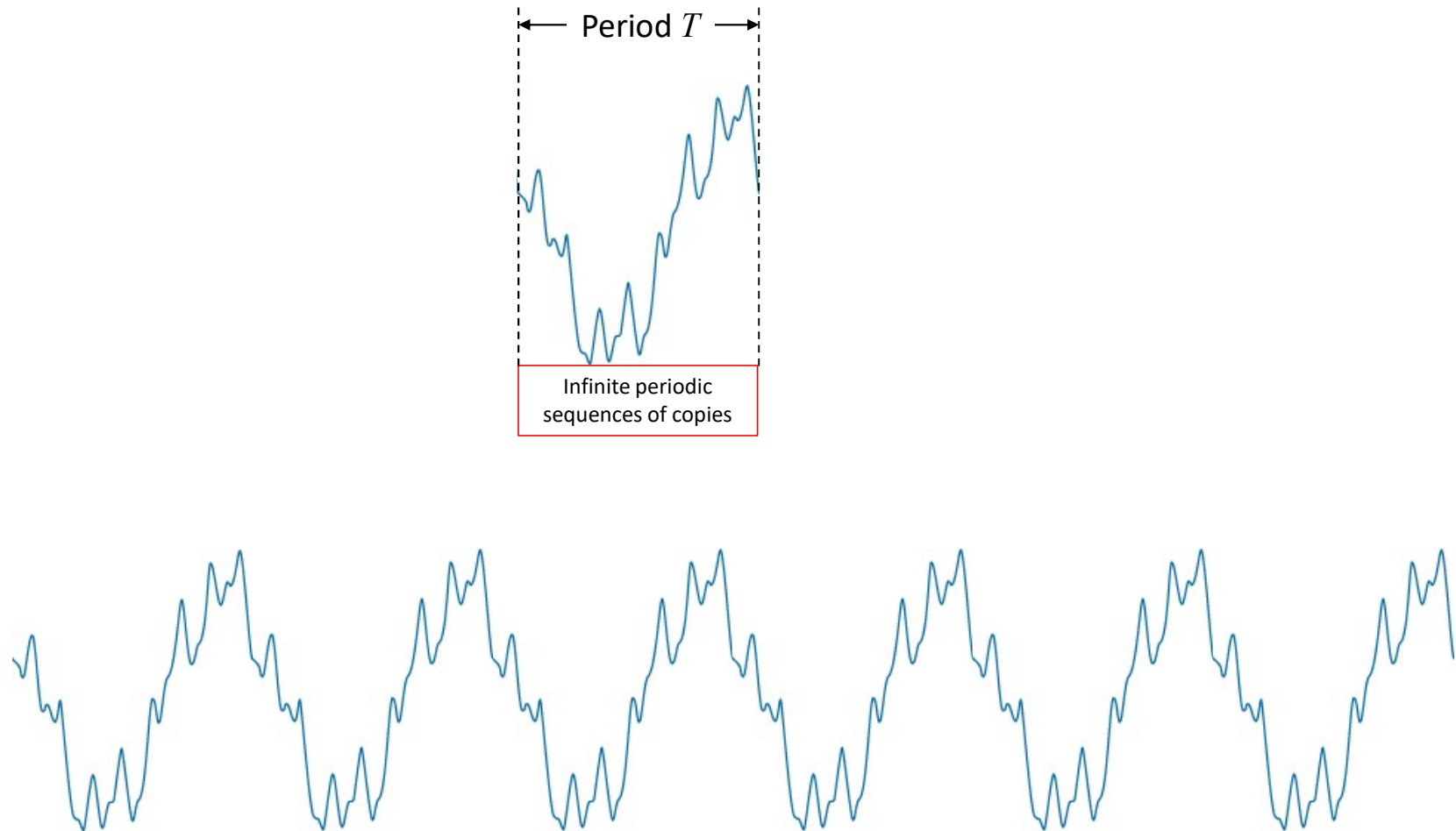
ECE 253 – Digital Image Processing

Filtering in the Frequency Domain

Overview: Image processing in the frequency domain



1D periodic functions



2D periodic functions



Fourier transform

1 D

(Forward) Fourier transform

$$F(\mu) = \int_{-\infty}^{\infty} f(t) e^{-j2\pi\mu t} dt$$

2 D

$$F(\mu, \nu) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t, z) e^{-j2\pi(\mu t + \nu z)} dt dz$$

Inverse Fourier transform

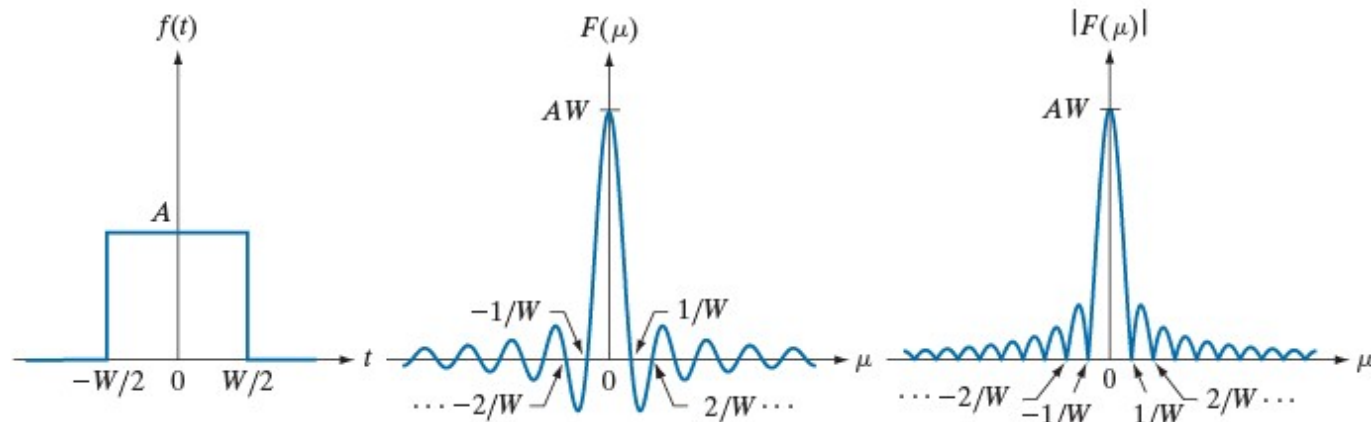
$$f(t) = \int_{-\infty}^{\infty} F(\mu) e^{j2\pi\mu t} d\mu$$

$$f(t, z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} F(\mu, \nu) e^{j2\pi(\mu t + \nu z)} d\mu d\nu$$

2D continuous Fourier transform

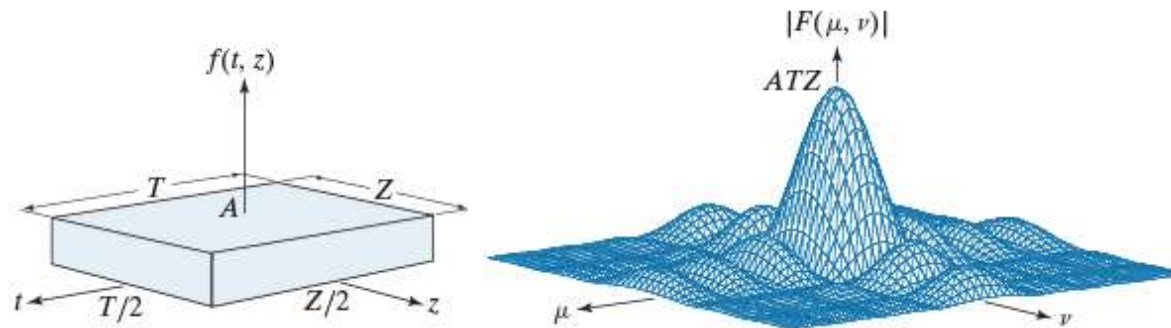
Example: box function

1D



The Fourier transform of a box function does not have an imaginary component (i.e., it is real)

2D



Impulse function (Discrete case)

1 D

Discrete variable x (at $x = 0$)

$$\delta(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{otherwise} \end{cases} \quad \left(\text{note } \sum_{x=-\infty}^{\infty} \delta(x) = 1 \right)$$

If $f(x)$ is discrete at $x = 0$, then

Unit impulse

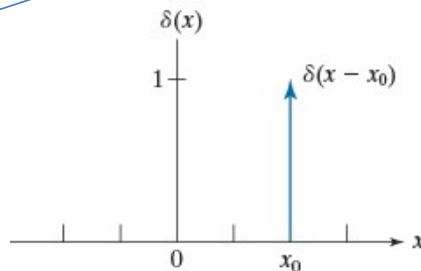
$$\sum_{x=-\infty}^{\infty} f(x) \delta(x) = f(0)$$

Generalizes to an arbitrary point x_0

Shifting property

$$\sum_{x=-\infty}^{\infty} f(x) \delta(x - x_0) = f(x_0)$$

The value of the function f at the location of the impulse (i.e., sample of the function)



2 D

Discrete variables x and y (at $x = y = 0$)

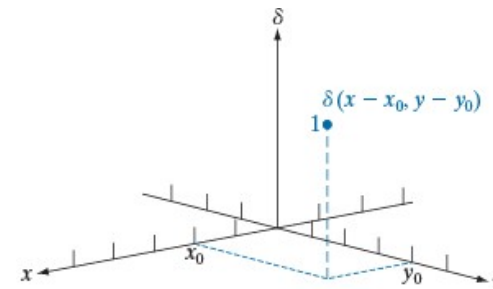
$$\delta(x, y) = \begin{cases} 1 & \text{if } x = y = 0 \\ 0 & \text{otherwise} \end{cases} \quad \left(\text{note } \sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} \delta(x, y) = 1 \right)$$

If $f(x, y)$ is discrete at $x = y = 0$, then

$$\sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x, y) \delta(x, y) = f(0, 0)$$

Generalizes to an arbitrary point (x_0, y_0)

$$\sum_{x=-\infty}^{\infty} \sum_{y=-\infty}^{\infty} f(x, y) \delta(x - x_0, y - y_0) = f(x_0, y_0)$$

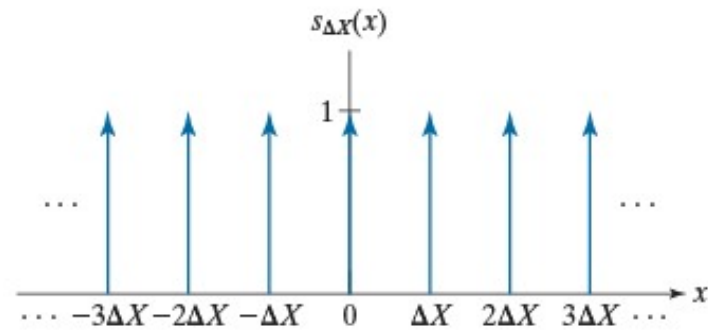
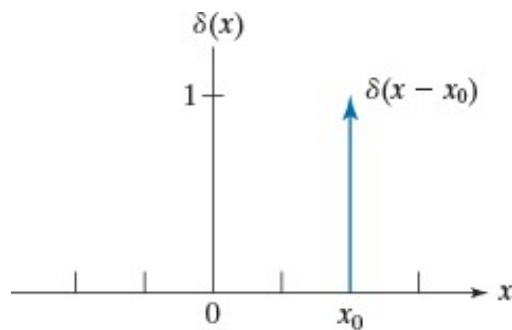


Discrete impulse function and impulse train

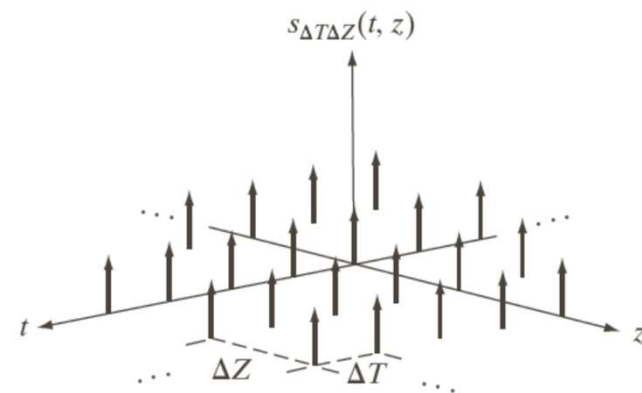
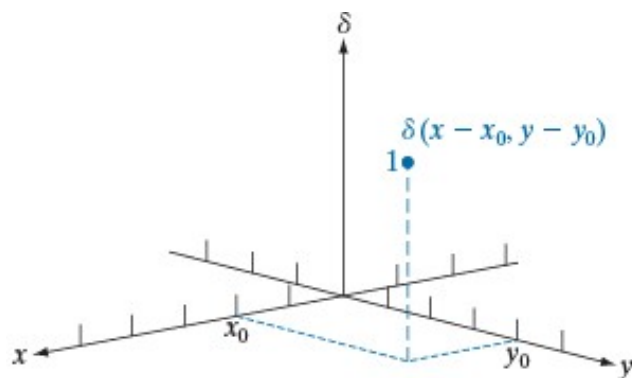
Impulse function

Impulse train

1D



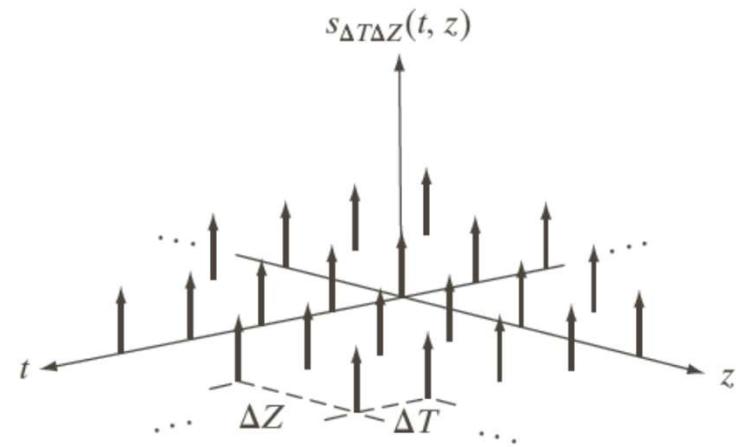
2D



2D discrete impulse train

Sum of infinitely many periodic impulses $\Delta T, \Delta Z$ units apart

$$s_{\Delta T \Delta Z}(t, z) = \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \delta(t - m\Delta T, z - n\Delta Z)$$

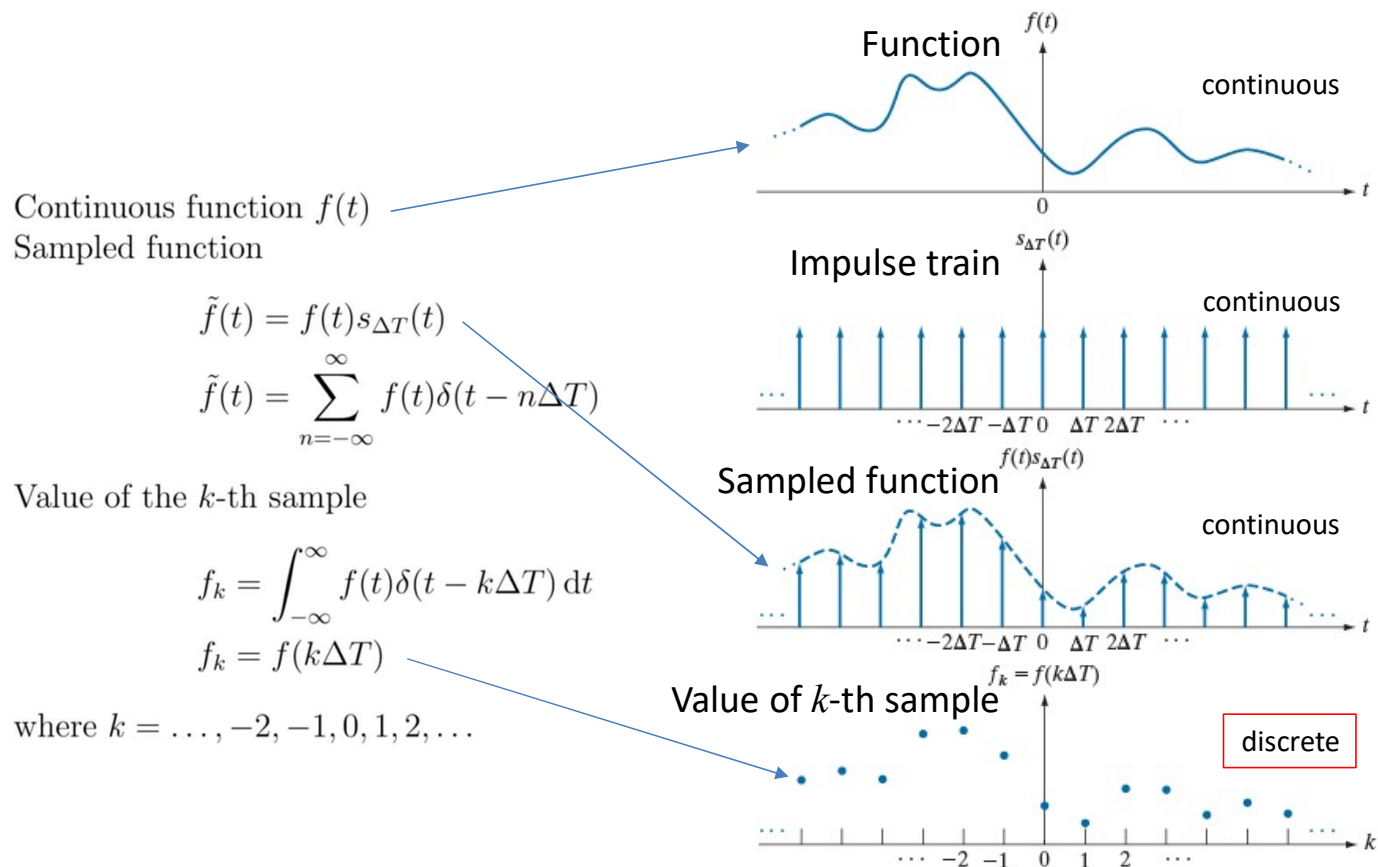


$f(t, z)$ can be recovered without error if

$$\Delta T < \frac{1}{2\mu_{\max}} \text{ and } \Delta Z < \frac{1}{2\nu_{\max}}$$
$$\frac{1}{\Delta T} > 2\mu_{\max} \text{ and } \frac{1}{\Delta Z} > 2\nu_{\max}$$

Guaranteed sufficient
separation between copies
No Aliasing

1D Sampling



1D Sampling – Frequency Domain Analysis

$$\tilde{F}(\mu) = \mathfrak{F} \{ \tilde{f}(t) \}$$

$$\tilde{F}(\mu) = \mathfrak{F} \{ f(t) s_{\Delta T}(t) \}$$

$$\tilde{F}(\mu) = F(\mu) \star S(\mu)$$

Convolution theorem

$$\tilde{F}(\mu) = \int_{-\infty}^{\infty} F(\tau) S(\mu - \tau) d\tau$$

$$\tilde{F}(\mu) = \frac{1}{\Delta T} \int_{-\infty}^{\infty} F(\tau) \sum_{n=-\infty}^{\infty} \delta \left(\mu - \tau - \frac{n}{\Delta T} \right) d\tau$$

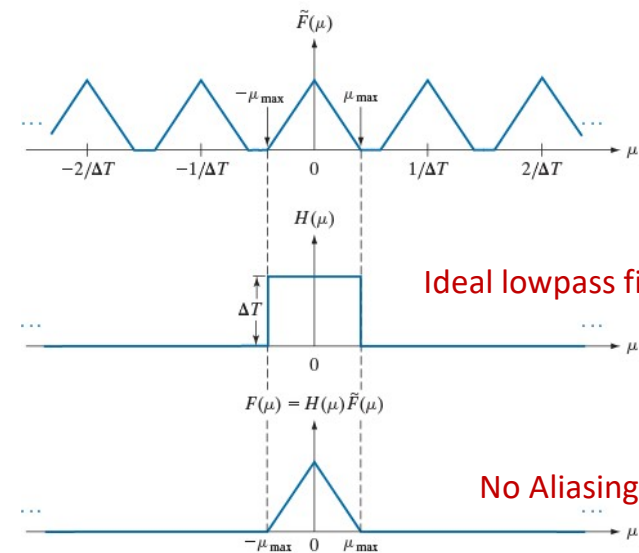
$$\tilde{F}(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} F(\tau) \delta \left(\mu - \tau - \frac{n}{\Delta T} \right) d\tau$$

Shifting property

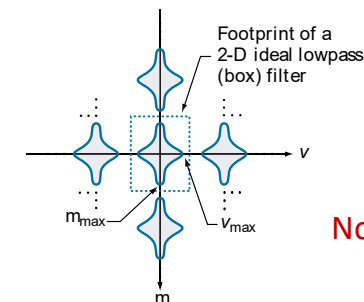
$$\tilde{F}(\mu) = \frac{1}{\Delta T} \sum_{n=-\infty}^{\infty} F \left(\mu - \frac{n}{\Delta T} \right)$$

This is an infinite, periodic sequence of copies of $F(\mu)$
 Since $F(\mu)$ is continuous, $\tilde{F}(\mu)$ is continuous

Fourier transform of **sampled** function



1D

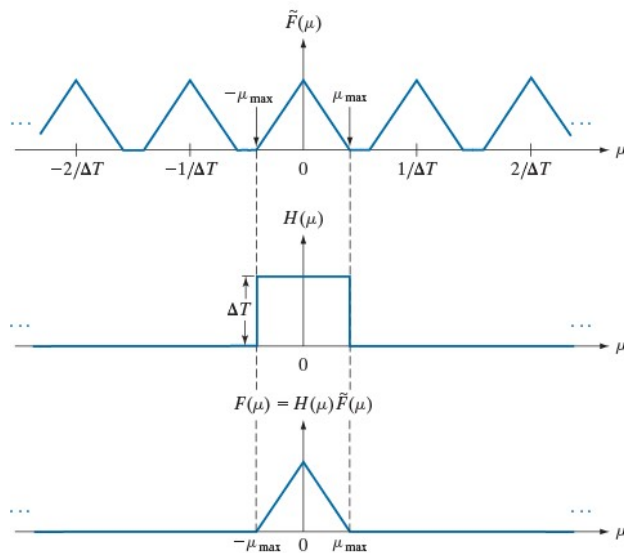


No Aliasing

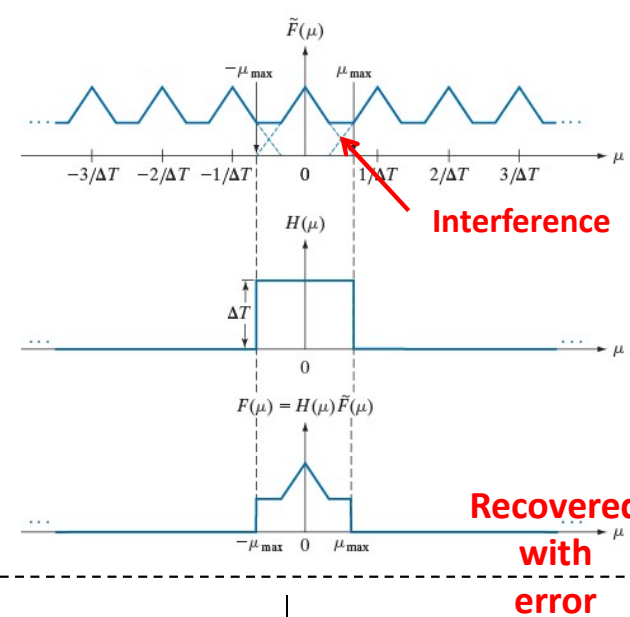
2D

Sampling – Frequency Domain Analysis

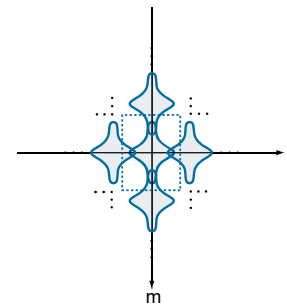
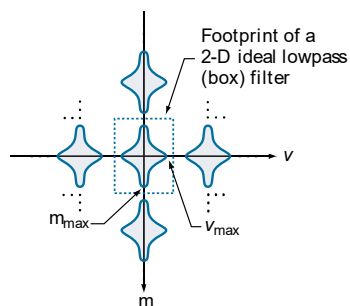
Oversampled. Recovered without error. No Aliasing.



Under-sampled. Recovered with error. Aliasing.

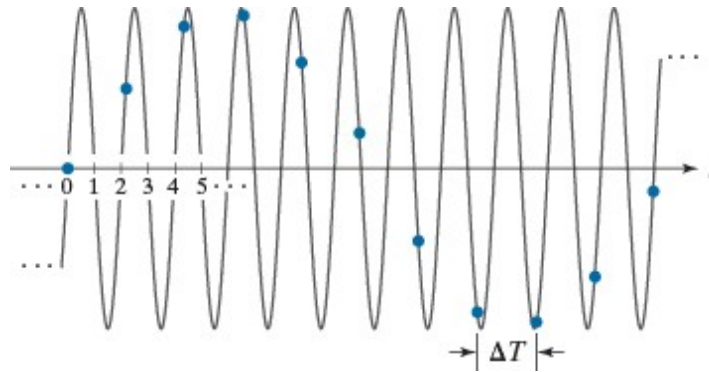


2D



Under-sampling → Aliasing

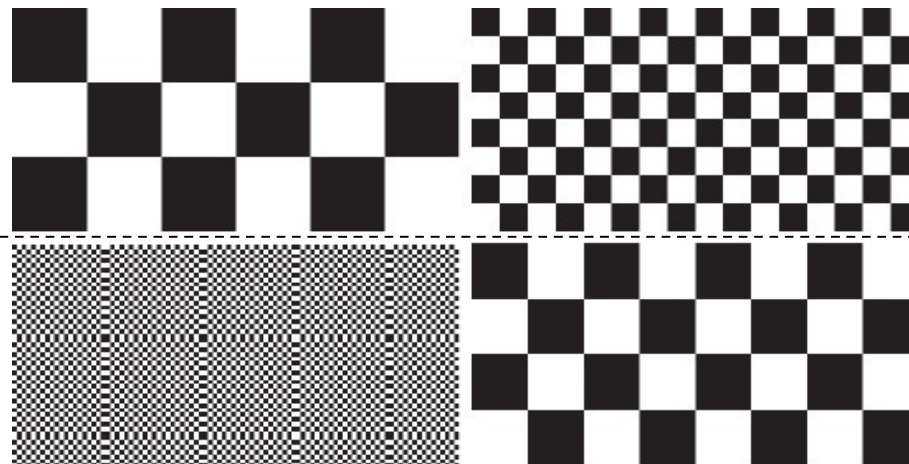
1D



Under-sampled
function looks
like a lower
frequency sine
wave

2D

In general, scaling
(zooming or
shrinking) an image
causes aliasing



Original

Aliasing

Aliasing in real images

Original

Aliasing

No aliasing



a b c

FIGURE 4.19 Illustration of aliasing on resampled natural images. (a) A digital image of size 772×548 pixels with visually negligible aliasing. (b) Result of resizing the image to 33% of its original size by pixel deletion and then restoring it to its original size by pixel replication. Aliasing is clearly visible. (c) Result of blurring the image in (a) with an averaging filter prior to resizing. The image is slightly more blurred than (b), but aliasing is not longer objectionable. (Original image courtesy of the Signal Compression Laboratory, University of California, Santa Barbara.)

Discrete Fourier transform (DFT)

1 D

2 D

(Forward) Fourier transform

$$F(u) = \sum_{x=0}^{M-1} f(x) e^{-j2\pi ux/M} \quad u = 0, 1, 2, \dots, M-1$$

Units of cycles/sample

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$u = 0, 1, 2, \dots, M-1$ and $v = 0, 1, 2, \dots, N-1$

Inverse Fourier transform

$$f(x) = \frac{1}{M} \sum_{u=0}^{M-1} F(u) e^{j2\pi ux/M} \quad x = 0, 1, 2, \dots, M-1$$

$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)}$$

$x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$

Each of these are an infinite, periodic sequence of copies

2D discrete Fourier transform (DFT)

Fourier transform pairs

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad u = 0, 1, 2, \dots, M-1 \text{ and } v = 0, 1, 2, \dots, N-1$$
$$f(x, y) = \frac{1}{MN} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)} \quad x = 0, 1, 2, \dots, M-1 \text{ and } y = 0, 1, 2, \dots, N-1$$

$$F(u, v) = \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad u = 0, 1, 2, \dots, M-1 \text{ and } v = 0, 1, 2, \dots, N-1$$
$$f(x, y) = \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)} \quad x = 0, 1, 2, \dots, M-1 \text{ and } y = 0, 1, 2, \dots, N-1$$

$$F(u, v) = \frac{1}{\sqrt{MN}} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)} \quad u = 0, 1, 2, \dots, M-1 \text{ and } v = 0, 1, 2, \dots, N-1$$
$$f(x, y) = \frac{1}{\sqrt{MN}} \sum_{u=0}^{M-1} \sum_{v=0}^{N-1} F(u, v) e^{j2\pi(ux/M + vy/N)} \quad x = 0, 1, 2, \dots, M-1 \text{ and } y = 0, 1, 2, \dots, N-1$$

2D discrete Fourier transform (DFT)

Zero-frequency term (DC term)

Note zero-frequency term is proportional to $\bar{f}(x, y)$, the average value of $f(x, y)$

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(ux/M + vy/N)}$$

$$F(0, 0) = MN \frac{1}{MN} \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y)$$

$$F(0, 0) = MN \bar{f}(x, y)$$

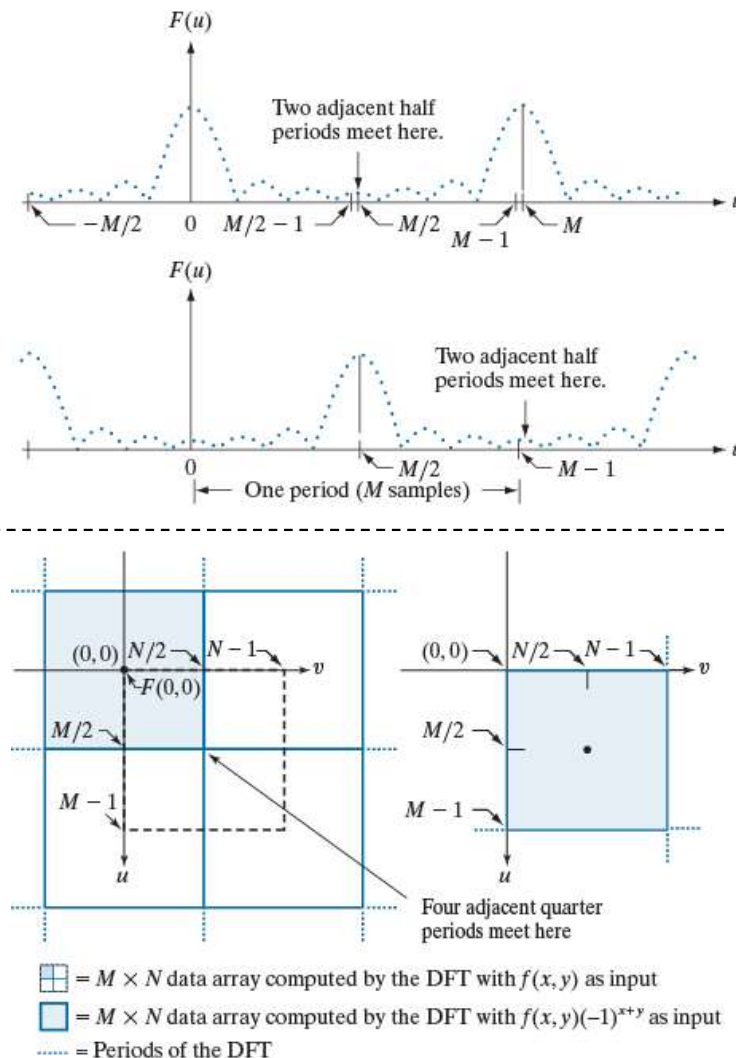
$$\|F(0, 0)\| = MN \|\bar{f}(x, y)\|$$

Centering the DFT

1D

Each of these are an infinite, periodic sequence of copies

2D



a
b
c d

FIGURE 4.22 Centering the Fourier transform. (a) A 1-D DFT showing an infinite number of periods. (b) Shifted DFT obtained by multiplying $f(x)$ by $(-1)^x$ before computing $F(u)$. (c) A 2-D DFT showing an infinite number of periods. The area within the dashed rectangle is the data array, $F(u, v)$, obtained with Eq. (4-67) with an image $f(x, y)$ as the input. This array consists of four quarter periods. (d) Shifted array obtained by multiplying $f(x, y)$ by $(-1)^{x+y}$ before computing $F(u, v)$. The data now contains one complete, centered period, as in (b).

Centering the DFT

Each of these are an infinite, periodic sequence of copies

Original

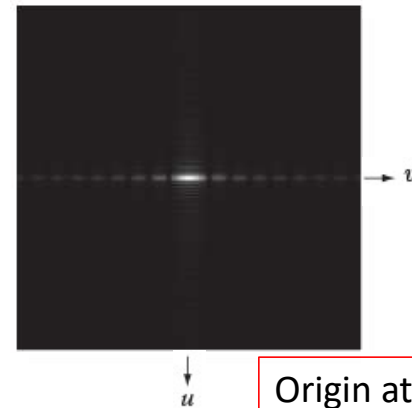


Origin at top left

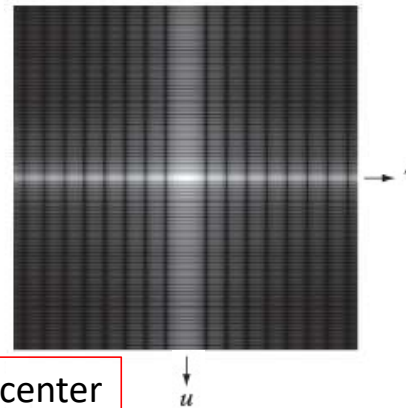


DFT magnitude
(look at corners)

Shifted
DFT magnitude



Origin at center

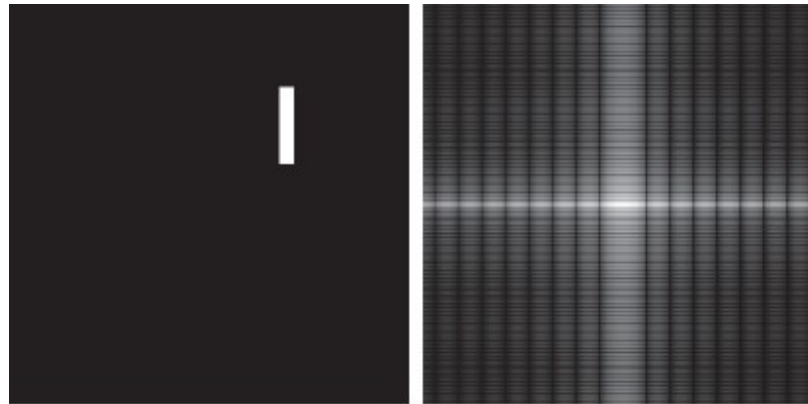


Log of shifted
DFT magnitude

DFT magnitude of geometrically transformed images

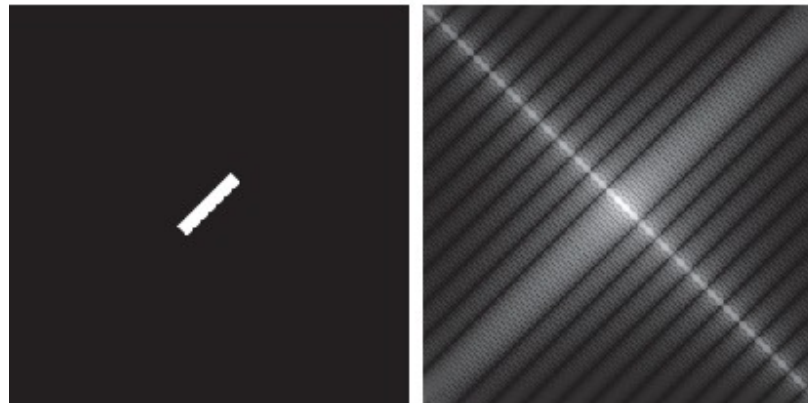
Each of these are an infinite, periodic sequence of copies

Translated



Same
magnitude
as original
(invariant to
translation)

Rotated
about
center



Different
magnitude
as original
(not invariant
to rotation)

DFT phase of geometrically transformed images

Each of these are an
infinite, periodic
sequence of copies

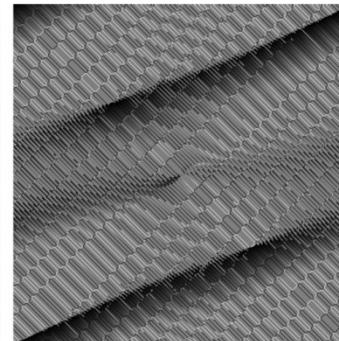
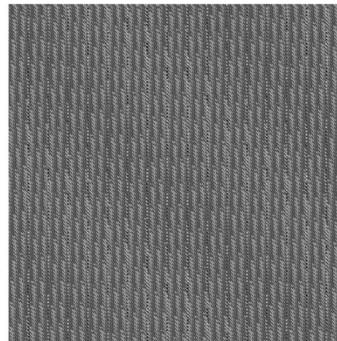
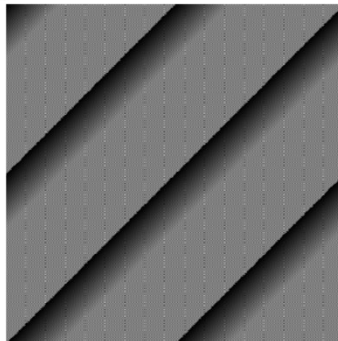
Original

Translated

Rotated
about center



DFT phase



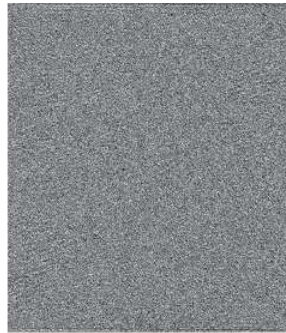
Contributions of magnitude and phase to image formation

Each of these are an infinite, periodic sequence of copies

Input image

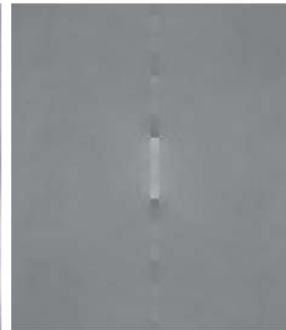


DFT Phase



Inverse DFT:
Phase only
(zero magnitude)

Inverse DFT:
Magnitude only
(zero phase)



Inverse DFT:
Boy magnitude
and
rectangle phase

Inverse DFT:
Rectangle magnitude
and boy phase

2D Convolution

2D discrete convolution

$$f(x, y) \star h(x, y) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f(m, n) h(x-m, y-n)$$

$x = 0, 1, 2, \dots, M-1$ and $y = 0, 1, 2, \dots, N-1$

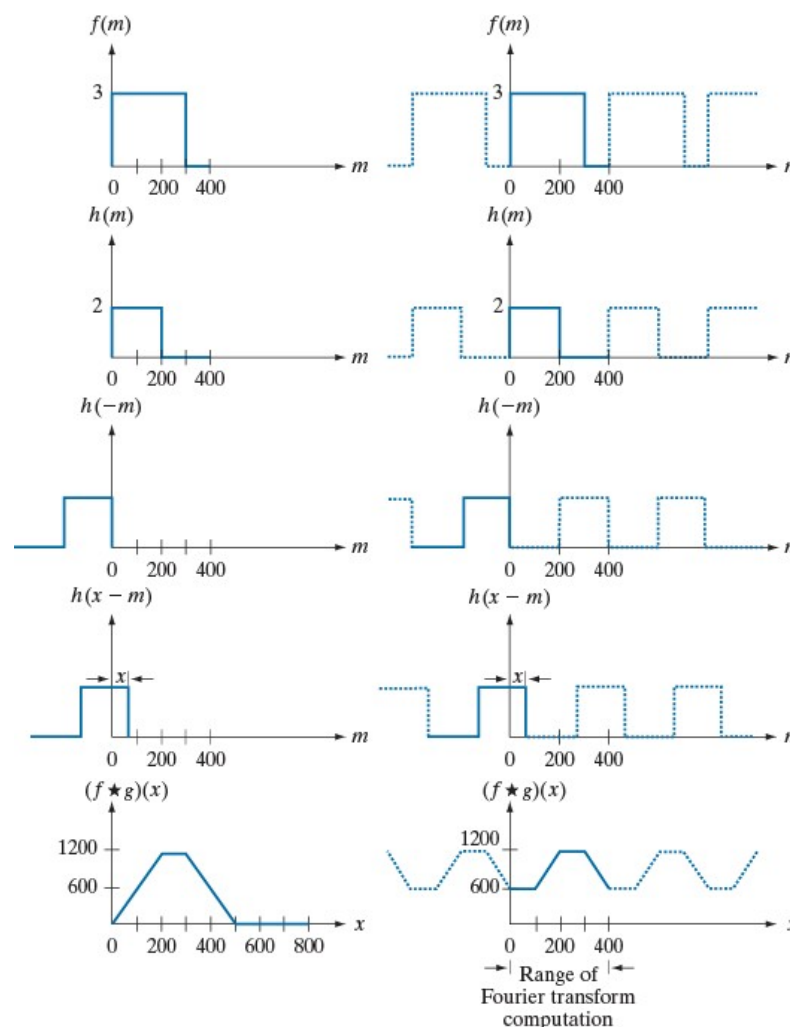
2D convolution theorem

$$f(x, y) \star h(x, y) \iff F(u, v) H(u, v)$$

$$f(x, y) h(x, y) \iff F(u, v) \star H(u, v)$$

Key to filtering in the frequency domain

Because the DFT is an infinite, periodic sequence of copies, the convolution is **circular**



Filtering in frequency domain using product without zero-padding

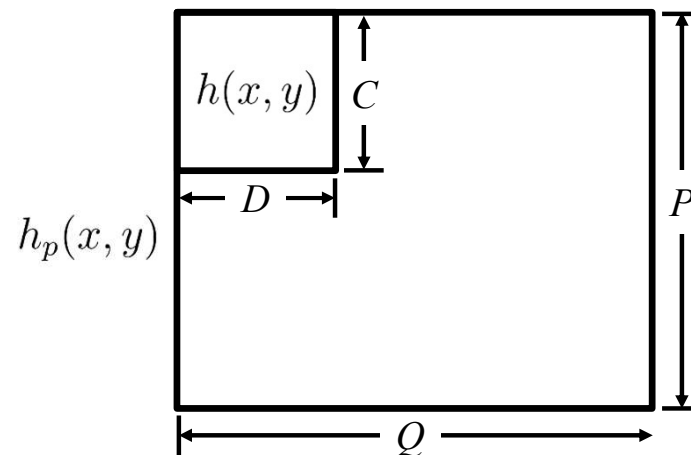
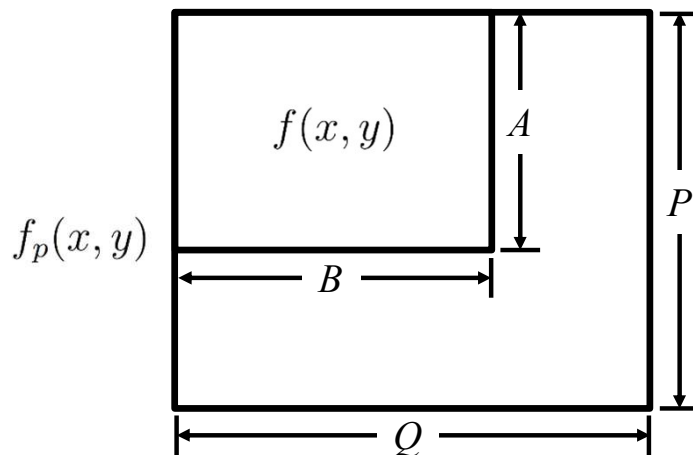
wraparound error

Filtering using convolution theorem

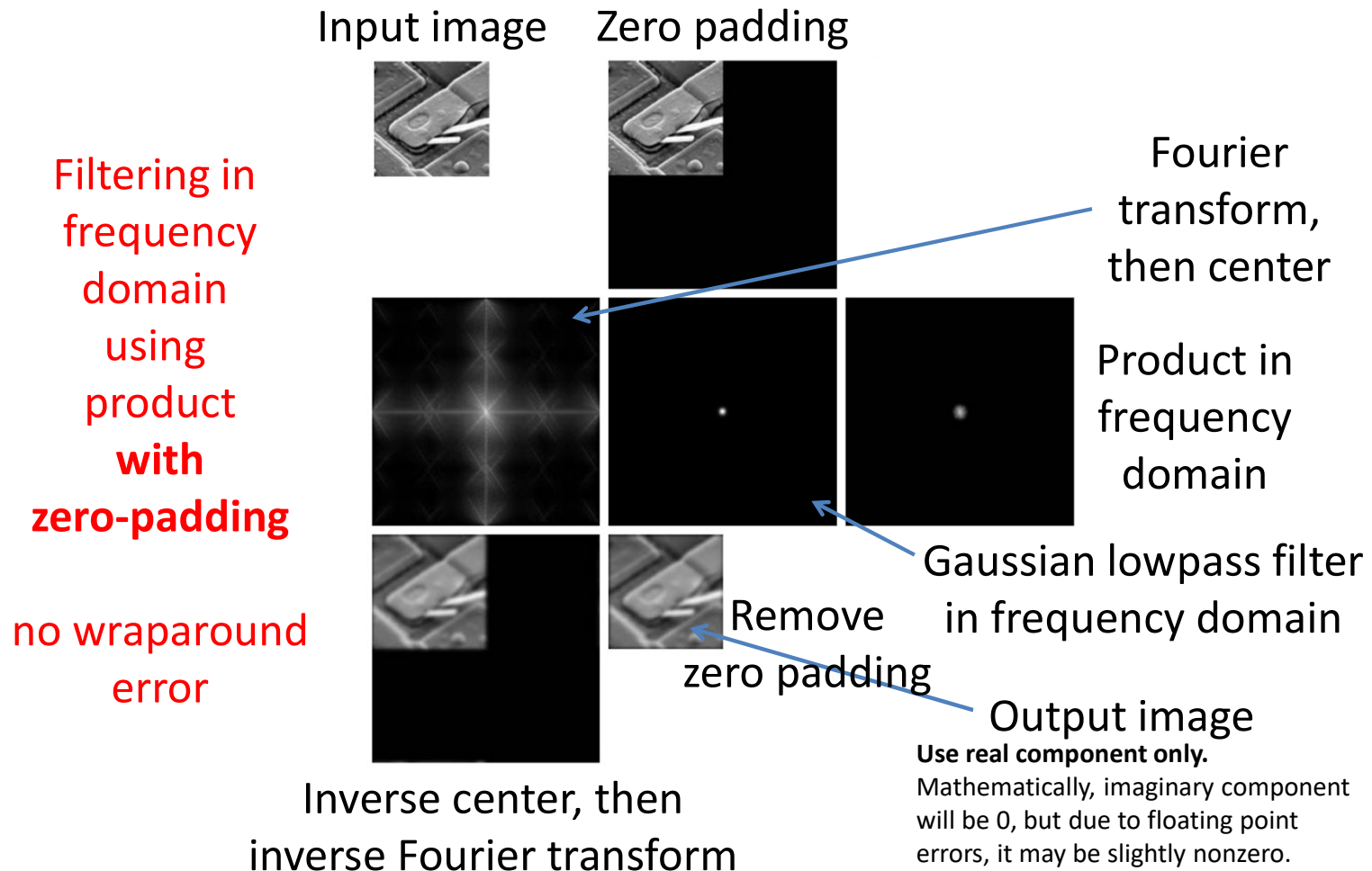
To avoid wraparound error, both functions must be zero padded

$$f(x, y) \star h(x, y) \iff F(u, v)H(u, v)$$

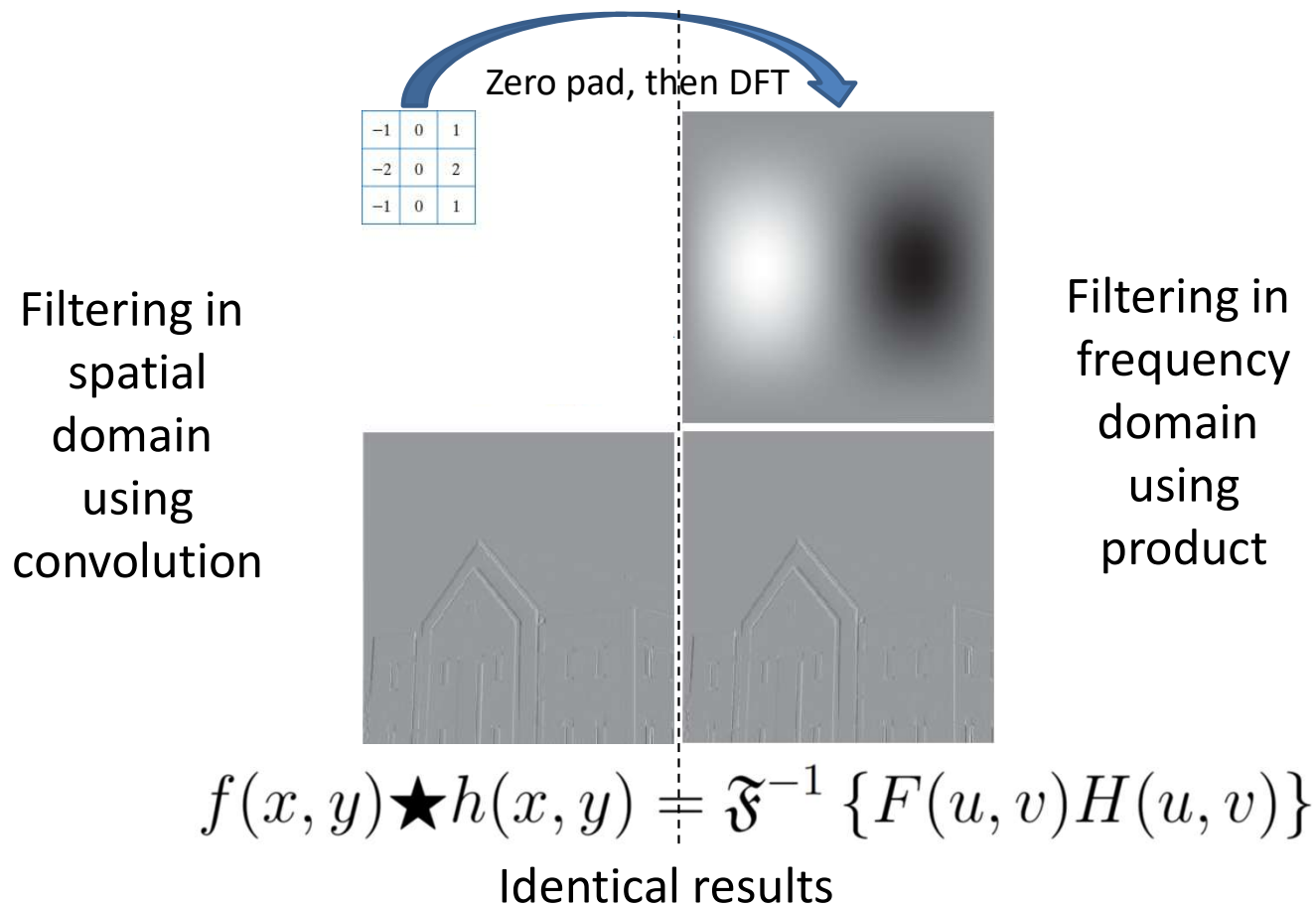
If $f(x, y)$ is $A \times B$ and $h(x, y)$ is $C \times D$, then both must be zero padded to $P \times Q$, where $P \geq A + C - 1$ and $Q \geq B + D - 1$



Filtering using convolution theorem



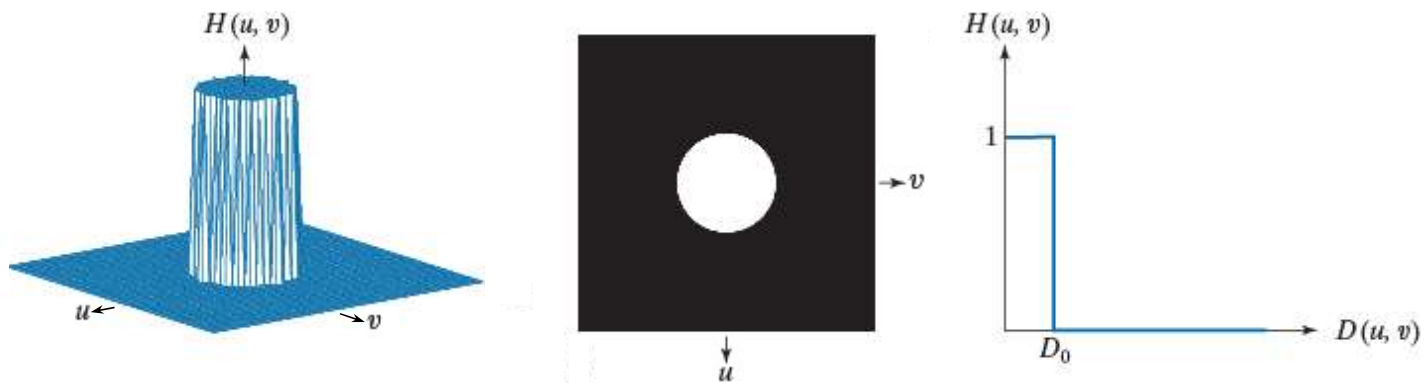
Filtering using convolution theorem



Filtering in the frequency domain

Ideal lowpass filter (LPF)

- Filter must be constructed in frequency domain



$$H(u, v) = \begin{cases} 1 & \text{if } D(u, v) \leq D_0 \\ 0 & \text{otherwise} \end{cases}, \text{ where } D(u, v) = \sqrt{\left(u - \frac{P}{2}\right)^2 + \left(v - \frac{Q}{2}\right)^2}$$

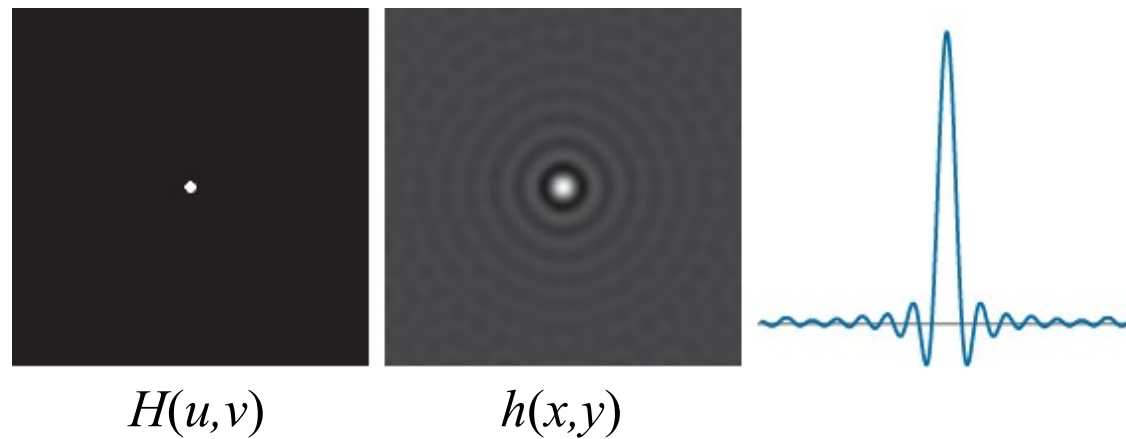
Distance from center of image

More accurately, use coordinates of $H(0,0)$ **after** centering

Filtering in the frequency domain

Ideal lowpass filter (LPF)

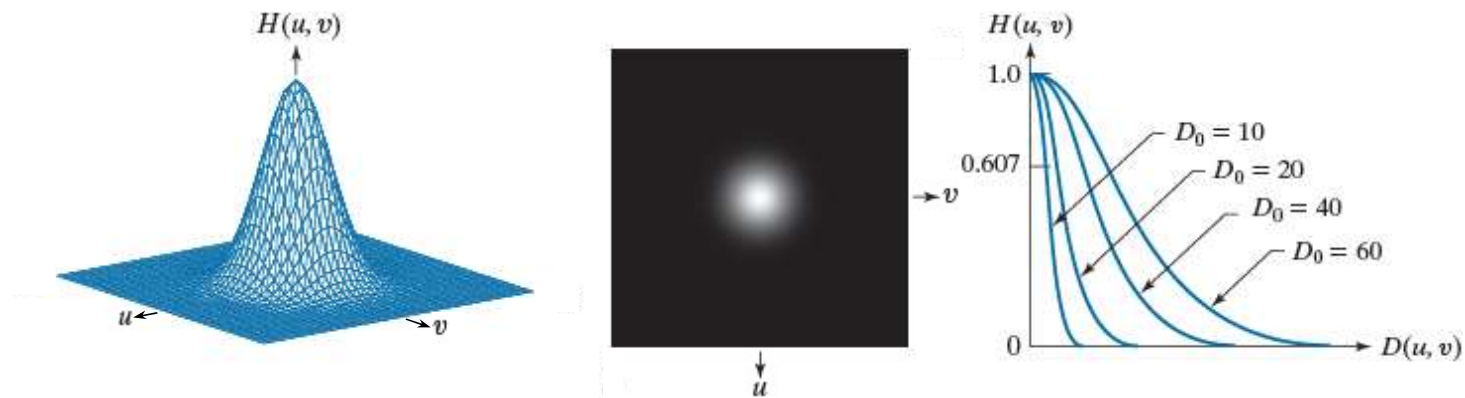
– Frequency domain vs spatial domain



Filtering in the frequency domain

Gaussian lowpass filter (LPF)

- Filter can be constructed in frequency domain directly

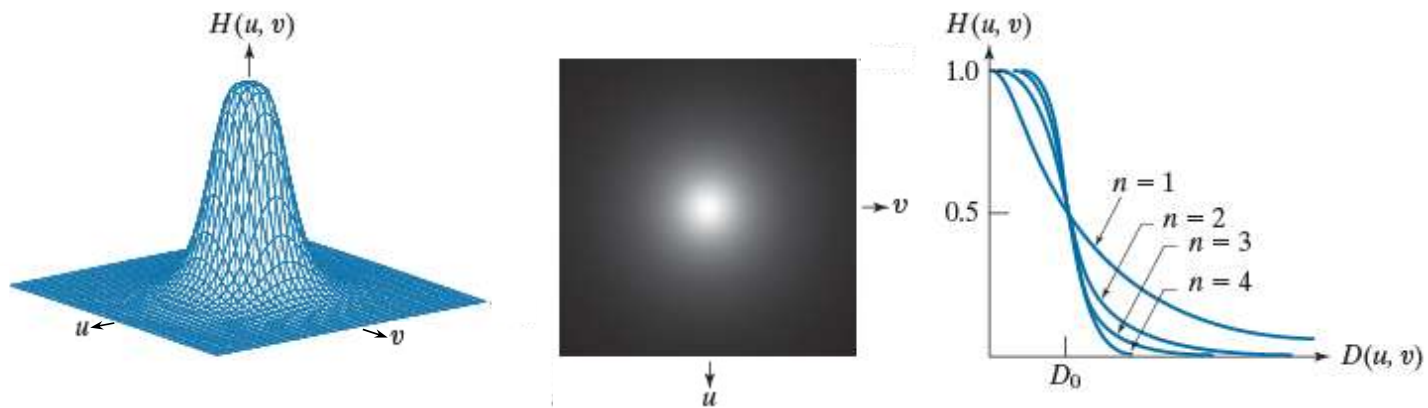


$$H(u, v) = e^{-D^2(u, v)/2\sigma^2}, \text{ where } D(u, v) \text{ is the distance from the center}$$

$$\text{Letting } \sigma = D_0, H(u, v) = e^{-D^2(u, v)/2D_0^2}$$

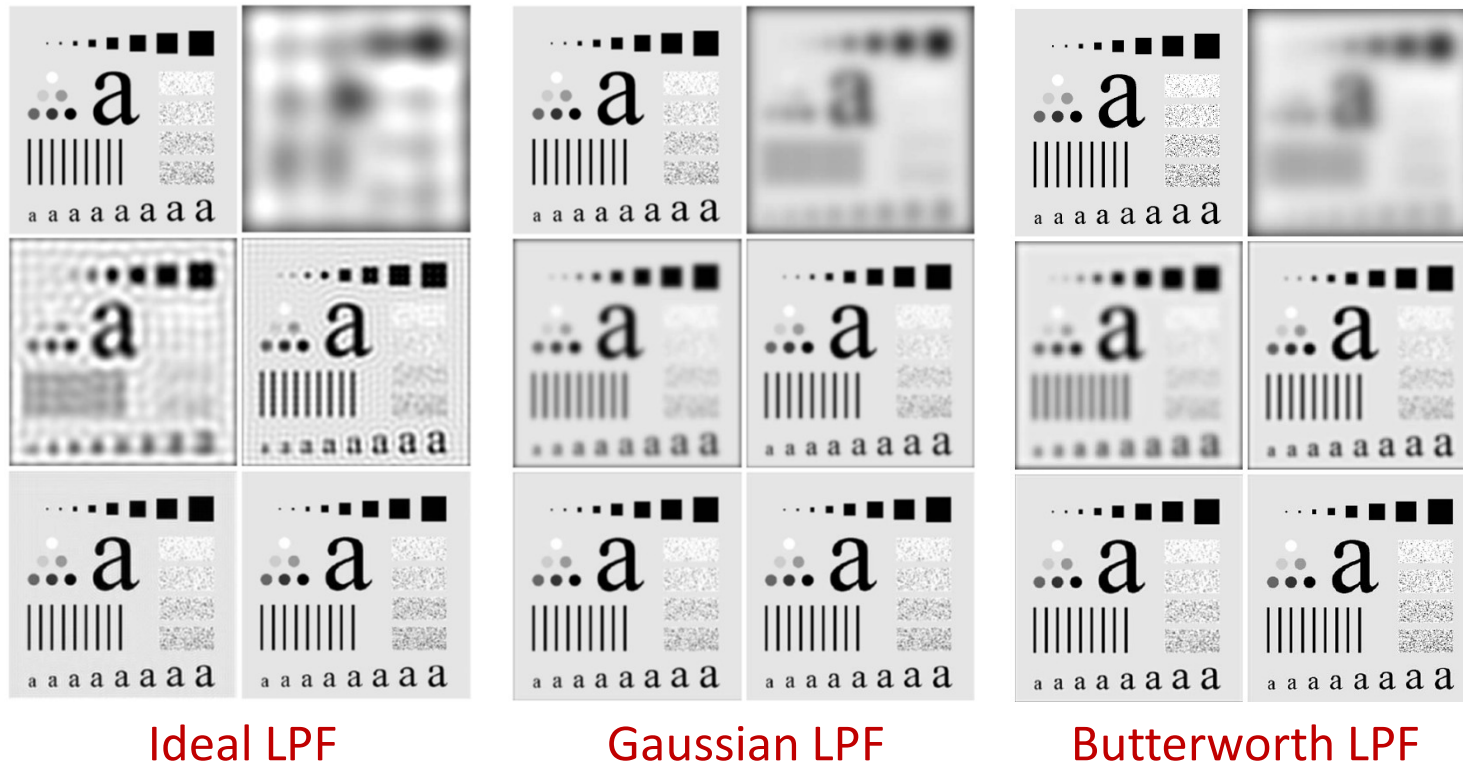
Filtering in the frequency domain

Butterworth lowpass filter (LPF)

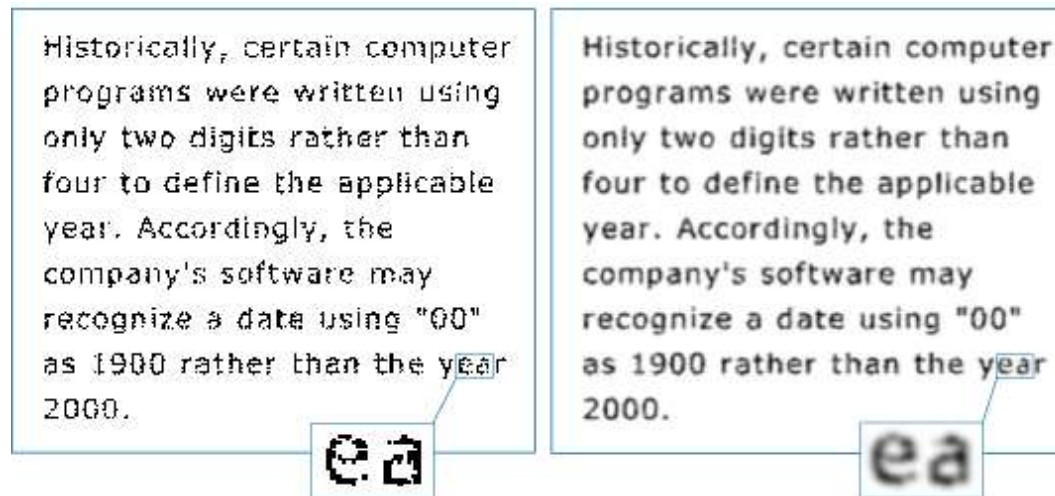


$$|H_b(jw)| = \frac{1}{\sqrt{1 + (w/w_c)^{2N}}}$$

Filtering in the frequency domain



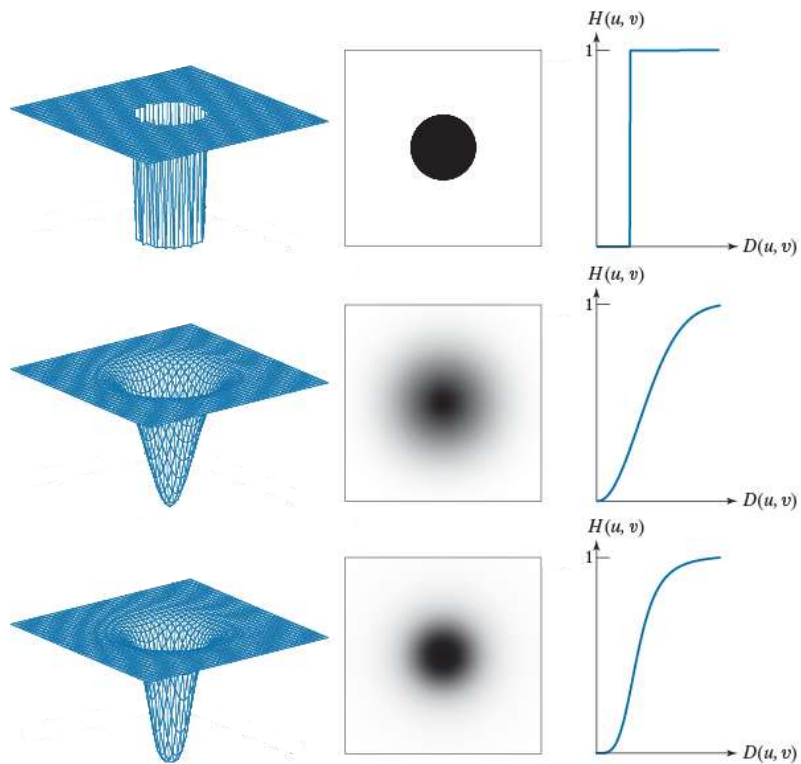
Example: character recognition



Gaussian LPF
joins broken characters

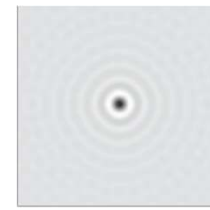
Highpass Filter (HPF)

Frequency Domain



Spatial Domain

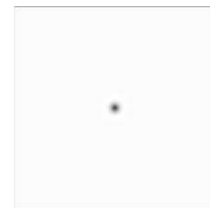
Ideal HPF



Gaussian HPF



Butterworth HPF

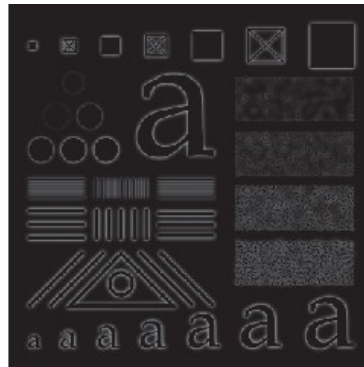


Filtering in the frequency domain

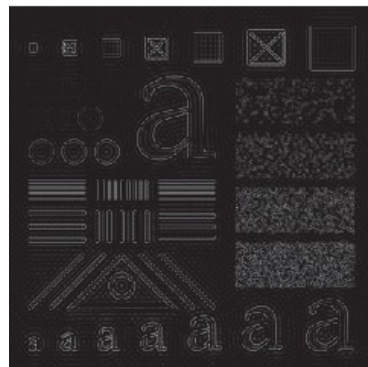
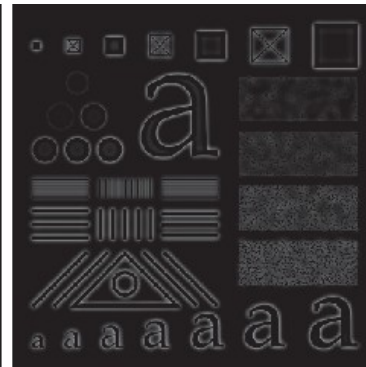
Ideal HPF



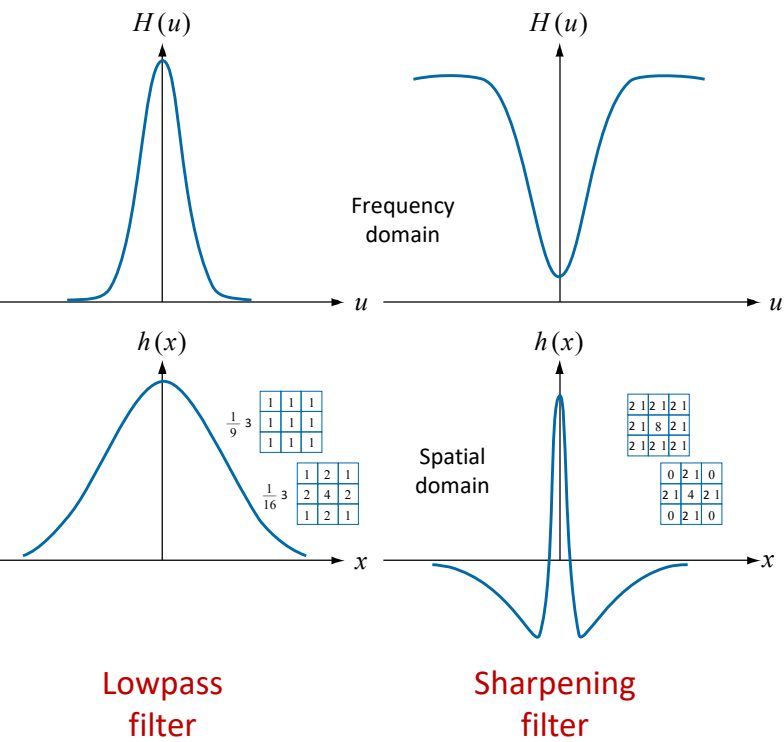
Gaussian HPF



Butterworth HPF



Filtering in the frequency domain



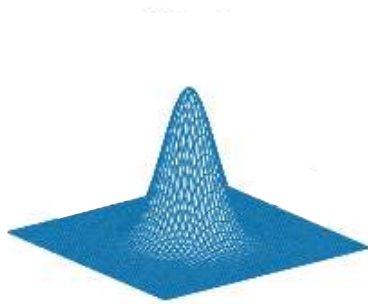
Original Image



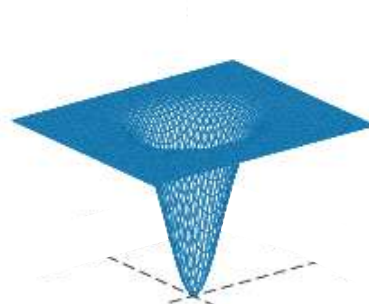
Sharpened Image

Filtering in the frequency domain

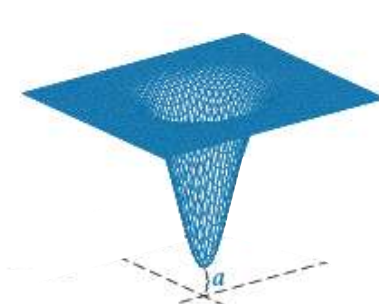
Lowpass filter



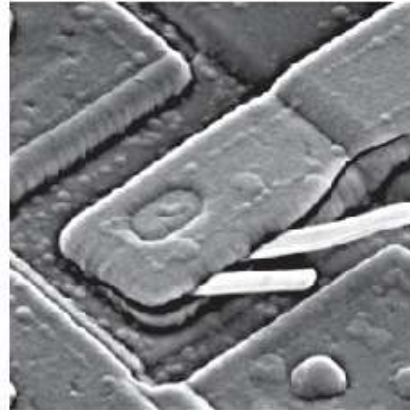
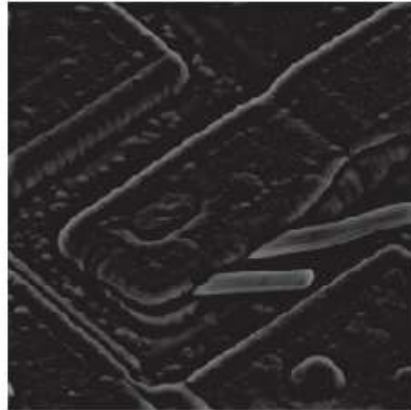
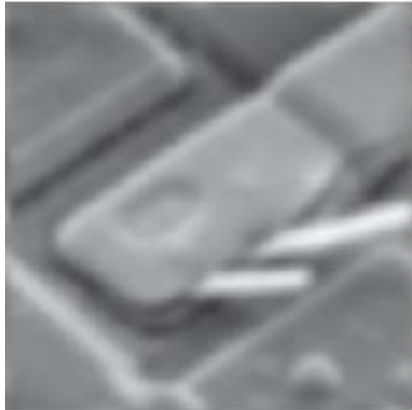
Highpass filter



Offset highpass filter



2D



Bandreject filters

Ideal

Gaussian

Butterworth

